Here is the Smart Learning Planner fourth-year final-year project end-to-end complete application flow.

1. The user opens the application.
2. The website will open with the landing page, a beautiful landing page having the login or registration button.
3. We will design the login page with the best frontend UI, whatever suits.
4. AI will provide me with ideas for this landing page and what content this landing page should hold.

   Next, the user authentication. We are using authentication via Clerk for login and registration. After clicking on the login or registration button, he will choose the role, whether he is a student or admin.
   Now, the student workflow:
5. The student profile setup will show on the first login only. He will set up his complete profile by selecting the semester, selecting the subjects, entering the learning goal, entering the exam dates, and entering the daily study availability.
6. The output will generate that the student profile will be created.

   Now, AI will create the initial study plan with the help of student information. AI will analyze it, and it will generate the personalized study plan, and a student dashboard opens.

   Next, now on the student dashboard. On the student dashboard page, a student can view:

- his study plan for today
- the upcoming tasks
- the study streak
- the calendar
- notifications
- the progress overview
- the week's topics
- AI recommendation

Now a student can access the different modules that we will implement.

Now, module number 1: study planner. The student will start today's task. The application will

show whether it is completed or not. Then Progress is updated.
Now, if a task is missed, then AI will adjust the future schedule accordingly.

Now, module number 2: The AI assistant. Now a student can ask the AI assistant to:

- create his revision plan
- explain a topic
- generate a quiz
- recommend resources
- show weak topics
- suggest today's priority now,

As soon as a student enters his query, the AI assistant will provide him with his personalized response. This personalized response will be perticular to the student.

Now, module number 3. The voice assistant
The voice assistant is just another input method for asking the query that will be used for the student. The student will provide the voice input of his query. Then the speech is converted into text and sent to the AI assistant. It will generate the personalized response, and the personalized response will be displayed.

Module number 4: learning progress as a student studies, the task, the quiz performance, study time, and consistency streak progress will be updated on his dashboard, and it will show on the dashboard of the student.

Now, module number 5 is the topic detection. The system or application will continuously check the quiz scores, revision frequency, incomplete topics, and study performances. According to these, weak topics will be identified, and a priority list will be generated.

Now, module number 6: it is a recommendation system. It will be done by using the learning data, learning history, weak topics, and progress. The AI will analyze all of these and recommend topics, videos, notes, practice questions, and the revision plan.

Now, module number 7: adaptive timetable. If student progress varies:
if the task is missed, or performance changes, then AI will reevaluate the schedule, and the updated timetable will be generated.

Now, module number 8 is learning analytics. The dashboard will display subject-wise progress, weekly performance, monthly performance, study time, completion rate, and learning trends.

Now, module number 9 is the notification system. The system sends reminders as:

- For upcoming exams, a study reminder will be sent.
- For upcoming deadlines, a task reminder will be sent.
- For week topics, a revision reminder will be sent.
- For schedule updates, a notification will be sent.

Now, module number 10: that is, resource discovery. The student will search for a topic. Thus, the system will find the best relevant resource, and the student starts learning from it. The progress of this learning will also be shown on his dashboard on a continuous basis.

Now, module number 11, that is, Profiles and Settings. It will include:

- Edit Profile or Change Password option
- Update Study Hours option
- Update Goal option
- Manage Notification Preferences option

Now, the semester learning cycle will be as follows:

1. Profile setup
2. AI will generate the study plan.
3. The student will study accordingly, on the basis of the dates assigned.
4. The progress will be tracked.
5. Topic detection will be done.
6. The AI recommendation system will do its work.
7. It will generate the additive timetable.
8. A revision plan will be generated.
9. The performance improvement metrics will be done.
10. If the goal is achieved, success or failure popups or notifications will show, and it will update the leaderboards.

Now, admin flow:

- Admin will log in via the application, and the admin dashboard will open.
- Admin can manage the students.

- Admin can manage the subjects.
- Admin can manage the learning resources.
- Admin can view the overall analytics.
- Admin can send the notifications.
- Admin can manage the system settings.

Now, a small application webflow is:
Now, the application will be opened. Then the login or registration button will show, and then the role selection will appear. There will be two roles: admin or student.

- Admin: If clicked on admin, it will open the dashboard. He can manage the system.
- Student: He will create his profile, and AI will create the plan. Then the student dashboard will appear. Then a study planner will be generated. Then the list of completed or incomplete tasks will show. Then progress tracking will be done. Then weak topic detection will be done. Then AI recommendation will be done. Then an edited timetable will be generated. Then learning analytics will be shown. Then a notification system will be done. Then resource discovery will be done. Then profiles and settings, where you can update and manage your profile. Then we can log out also.